Problem 0

(a) Sketch the region $x > y^2 - 1$.

(b) Sketch the region $y > x^2 + 2x + 1$.

(c)* **(hard / bonus / only if we have time)**

Find the area of overlap between these two regions.

Problem 1

(a) Classify each surface as a cylinder, paraboloid, hyperboloid, ellipsoid, sphere, or none of the above.

(i) $z = 2x^2 + 3y^2$ _____

(ii) $x^2 + 4y^2 + 9z^2 = 36$ _____

(iii) $x^2 + y^2 + z^2 - 2x + 4y - 6z = 2$ _____

(iv) $x^2 + y^2 - z^2 = 1$ _____

(v) $z = (x - 1)^2$ _____

(vi) $z^2 = x^2 + y^2$ _____

Problem 1 (continued)

(b) Match each function to its surface below. Each graph is used exactly once.

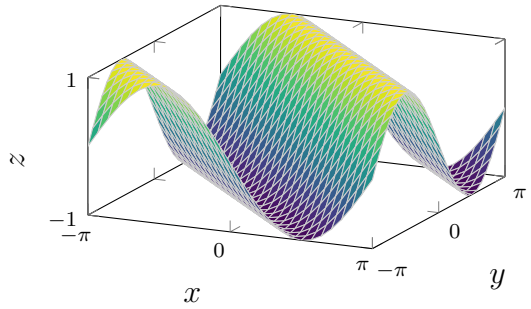
(i) $f(x,y) = \dfrac{1}{1+x^2+y^2}$ (ii) $g(x,y) = 1 - x^2 - y^2$

(iii) $h(x,y) = \dfrac{1}{1+(x^2+y^2-2)^2}$ (iv) $k(x,y) = \cos\left(2\sqrt{x^2+y^2}\right)$

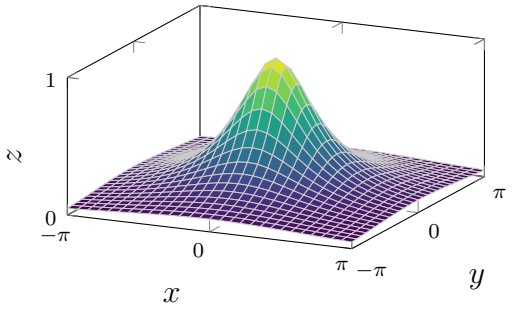
(v) $p(x,y) = \sin x \sin y$ (vi) $q(x,y) = \sin(x+y)$

All plots show $-\pi \le x, y \le \pi$ from the same viewpoint.

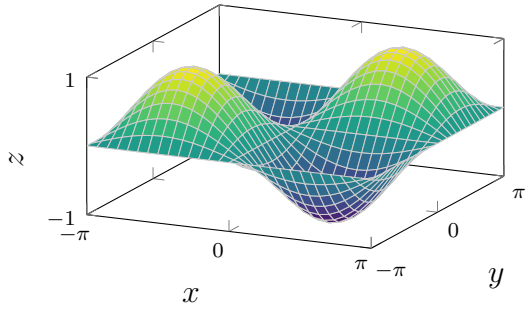
A



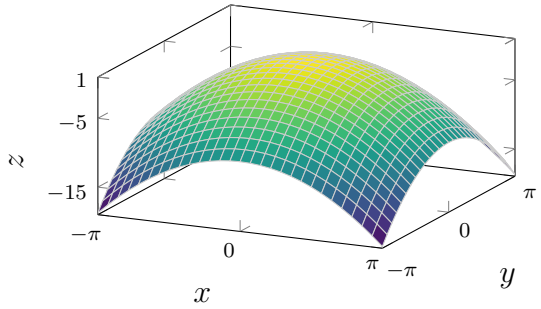
B



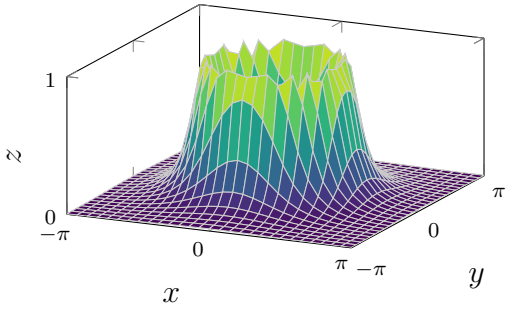
C



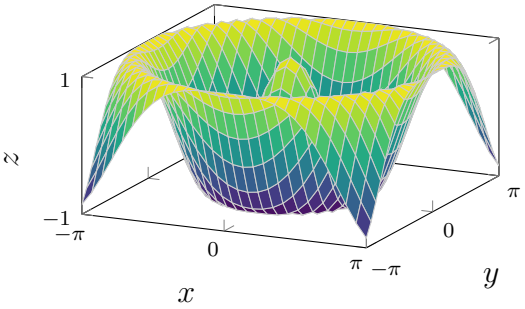
D



E

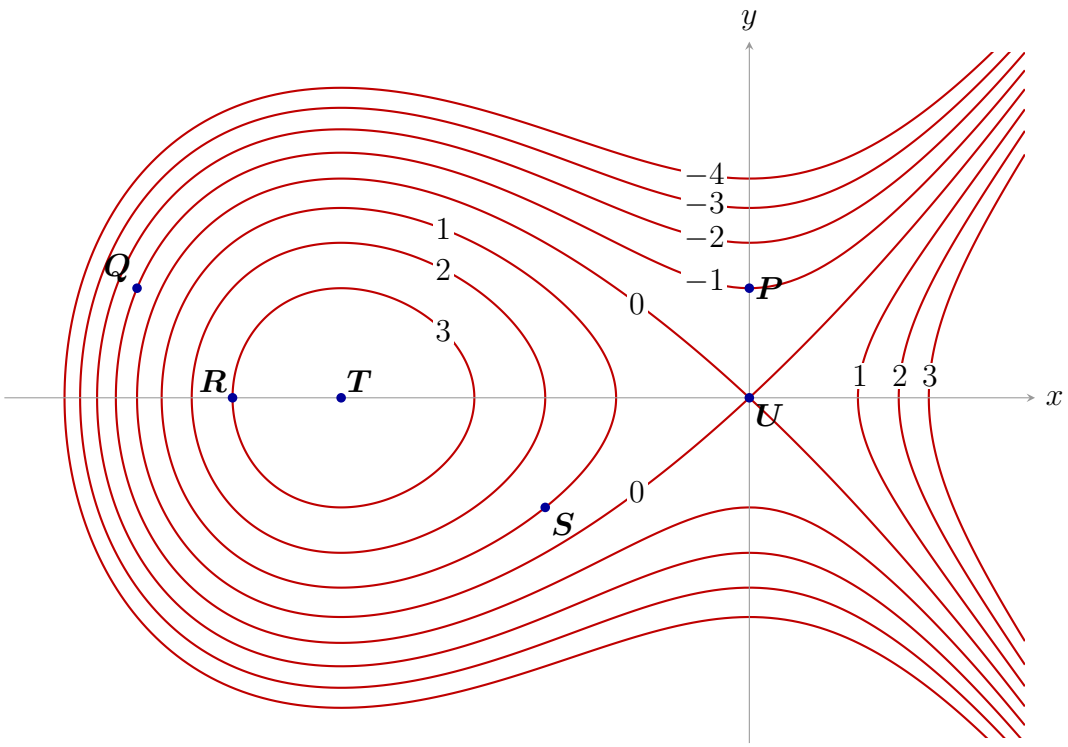


F



Problem 2

The diagram shows level curves of a smooth function $f(x, y)$. For parts (a)–(d), circle all labeled points where the statement applies. Assume that f has a local maximum at T.



(a) $\nabla f = \mathbf{0}$. $P \quad Q \quad R \quad S \quad T \quad U$

(b) f has a saddle point. $P \quad Q \quad R \quad S \quad T \quad U$

(c) $f_y > 0$. $P \quad Q \quad R \quad S \quad T \quad U$

(d) The directional derivative along $\langle 0, -1 \rangle$ is negative. $P \quad Q \quad R \quad S \quad T \quad U$

(e) Draw the gradient at each labeled point. Which point has the largest gradient magnitude?

Problem 3

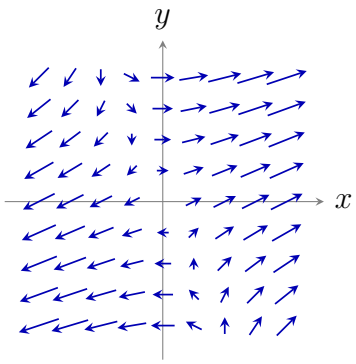
(a) Match each vector field to a plot below.

(i) $\langle -y, x \rangle$ (ii) $\langle x - y^2, y \rangle$ (iii) $\langle x + y, x + y \rangle$ (iv) $\langle y, x \rangle$

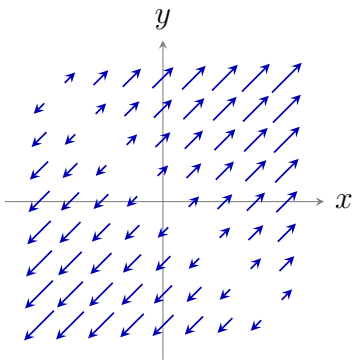
(b) Match each function to the plot of its gradient.

(i) $\frac{1}{2}(x^2 + y^2)$ (ii) xy (iii) $(x + y)^2$ (iv) $x(x + y)$

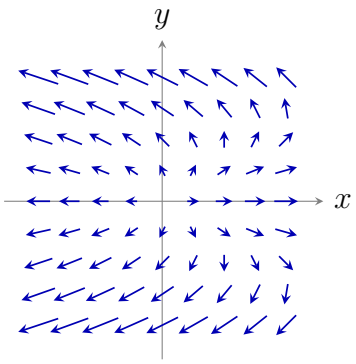
A



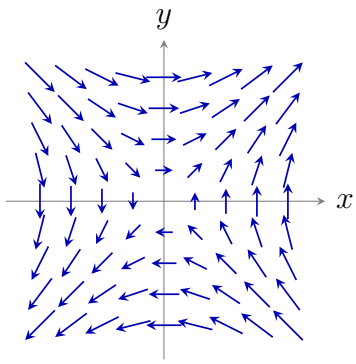
B



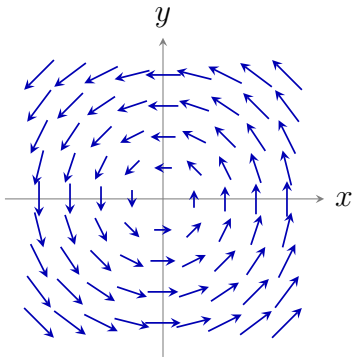
C



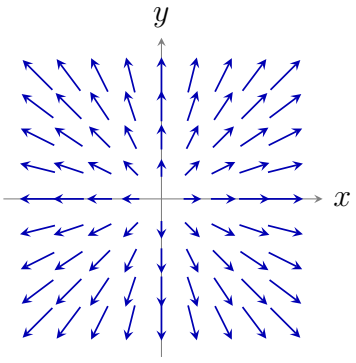
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F



(c) Draw the unit circle, *positively oriented*, on each plot. (If you haven't covered this in class, it means that the interior of the region is on your left as you travel around the curve.) For each field, is its vector line integral around this circle positive, negative, zero, or hard to tell just from looking?

Problem 4

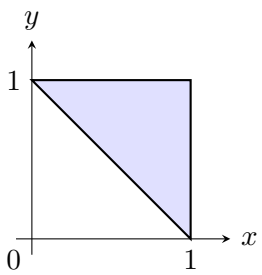
(a) Match each integral to its region below. Do not evaluate.

(i) $\displaystyle\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} dx\, dy$ (ii) $\displaystyle\int_0^1 \int_{x^2}^x dy\, dx$ (iii) $\displaystyle\int_0^1 \int_{1-y}^1 dx\, dy$

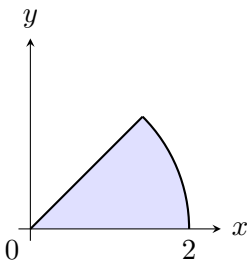
(v) $\displaystyle\int_0^1 \int_{\sqrt[3]{1-y}}^1 dx\, dy$ (vi) $\displaystyle\int_0^1 \int_0^{x^3} dy\, dx$

(iv) $\displaystyle\int_{-2}^{-1} \int_0^{\sqrt{4-x^2}} dy\, dx + \int_{-1}^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} dy\, dx + \int_1^2 \int_0^{\sqrt{4-x^2}} dy\, dx.$

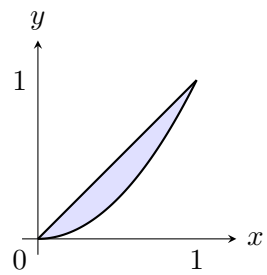
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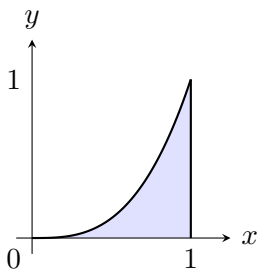
B



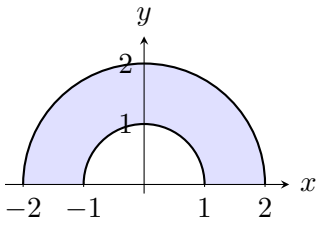
C



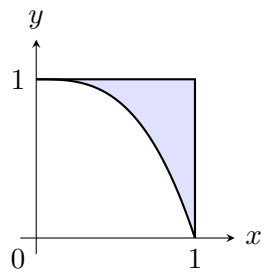
D



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(b) Rewrite each integral in a different (preferably simpler) order. Do not evaluate.